STRUCTURED AND OBJECT ORIENTED PROGRAMMING

PROJECT REPORT

"USSA STUDENT PORTAL"

Course Unit: CSC 1102 – Structured and Object-Oriented Programming

GROUP 23

Class: DAY

Group Leader: OCEN ANDREW DESIRE

GROUP MEMBERS

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GitHub Repository Link: https://github.com/TerryStunner/GROUP-23-PROJECT-

GROUP 23 PROJECT DOCUMENT

1. Introduction

The USSA Student Portal is an application developed to provide students with a central platform for accessing academic information and selected educational services.

The system brings together different learning services, including live lessons, recorded lessons, learning resources, payment reference recording and student discussions.

2. Objectives of the Application

To provide students with a central platform for accessing learning information, live and recorded lessons, access to educational resources like text books, websites and past papers, provide a simple interface for recording payment references and encourage student interaction through a discussion section.

3. Main Features of the Application

3.1 Student Welcome and Entry

The application features a welcome screen where a student enters their name before accessing the portal. The system checks that the name field is not empty before proceeding.

3.2 Home Section

The home section welcomes the student and provides an overview of the USSA Student Portal and its available sections.

3.3 Live Lessons

This section displays scheduled online lessons. Students can view: Course units, Lesson titles, Presenters, Lesson schedules, Lesson descriptions, and Online meeting links.

The application also provides options for viewing lesson details and opening the associated links.

3.4 Recorded Lessons

This section provides access to recorded learning sessions. Students can browse available recordings, view descriptions and open the associated video links.

3.5 Learning Resources

The resources section provides students with learning materials and other educational resources. Each resource contains a title, description, resource type and associated link.

3.6 Payments Section

The payments section allows students to select a payment plan, such as daily, weekly or monthly. Students can enter a transaction reference for recording. The application displays the payment status as pending verification.

3.7 Student Discussions

This section allows students to enter and post discussion messages. Messages are displayed within the application during the current session.

3.8 About and Help

This section provides information about the academy and basic guidance on using the application.

4. Classes Used in the Application

The application consists of six main Java classes.

4.1 USSAApp Class

This is the main application class. It creates the main application window, Implements the Java Swing graphical interface.

Creates and manages application tabs, learning content, payment references, discussions, and execution control.

4.2 Student Class

This class represents a student using the application.

Stores student information (name, ID).

Provides constructors, getters, setters, and detail retrieval methods.

4.3 LearningContent Class

This is an abstract parent class representing learning materials.

Stores common content info, provides constructors/getters, defines description methods, and declares abstract getContentType().

4.4 LiveLesson Class

This class extends LearningContent. It represents scheduled online lessons and overrides getContentType() for live lessons.

4.5 RecordedLesson Class

This class extends LearningContent. It represents recorded lessons and provides the content type for recorded materials.

4.6 Resources Class

This class extends LearningContent and represents educational resources available to students.

5. How the Classes Interact

The USSAApp class acts as the main controller of the application. It is responsible for creating and managing the different learning-content objects, which include LiveLesson, RecordedLesson, and Resources. All these objects are stored in one array of type LearningContent[].

Since the three classes inherit from LearningContent, they can all be accessed using the parent class. This makes it easier to manage the different types of learning content in one place instead of having to create a separate array for each one.

Even though they are stored together, each class still has its own information and purpose. For example, a live lesson has details about its scheduled time, while a recorded lesson provides information about previously recorded sessions.

When a user selects a particular item, the application retrieves and displays the relevant information. This allows the different types of learning content to be managed together while still keeping their individual features.

6. Object-Oriented Programming Concepts Demonstrated

6.1 Encapsulation

Uses private attributes and public getters/setters in the Student and content classes to control data access.

6.2 Inheritance

LiveLesson, RecordedLesson, and Resources inherit common attributes and methods from LearningContent.

6.3 Abstraction

LearningContent is an abstract class defining a unified structure for all sub-content types.

6.4 Polymorphism

Objects stored under LearningContent call overridden methods like getContentType() for dynamic behavior.

6.5 Constructors

Used to initialize objects upon instantiation.

7. Graphical User Interface

The application uses Swing to develop its graphical user interface. Key components used:

Component	Purpose
JFrame	Creates the main application window.
JPanel	Organises interface components.
JTabbedPane	Organises application sections into tabs.
JLabel	Displays text and headings.
JTextField	Accepts user input.
JButton	Allows users to perform actions.

Component	Purpose
JList	Displays lists of learning content.
JComboBox	Allows selection of payment plans.
JTextArea	Displays and accepts longer text.
JOptionPane	Displays messages and alerts.
JScrollPane	Provides scrolling for displayed information.

The application uses event listeners to respond to user actions such as button clicks and text submissions.

8. Use of Arrays

LearningContent[]: Stores live lessons, recorded lessons, and resources.

String[] paymentRecords: Stores payment references entered during the active session.

String[] discussions: Stores discussion messages posted during the session.

9. Flow Control

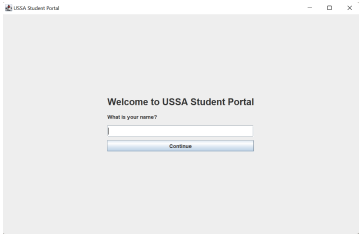
Conditional Statements

if/else statements validate student names, payment entries, discussion messages, and prevent invalid operations.

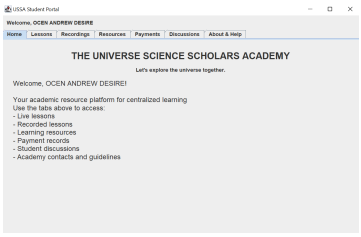
Iterative Statements

For-each and indexed for loops count items, display learning materials, filter content by type, and display discussion posts. Array capacity is checked prior to additions.

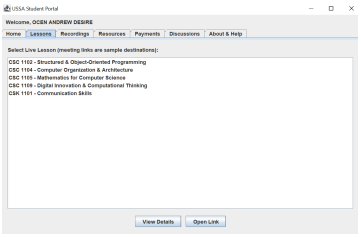
10. Application Screenshots



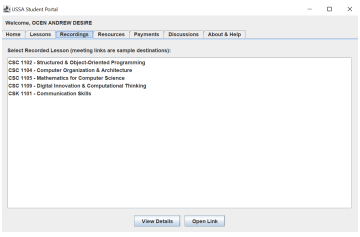
Welcome Screen



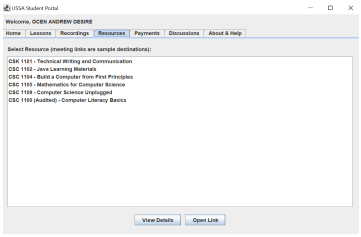
Home Screen



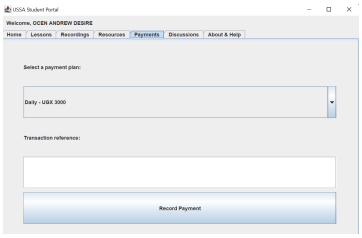
Live Lessons



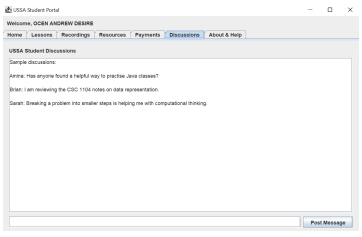
Recorded Lessons



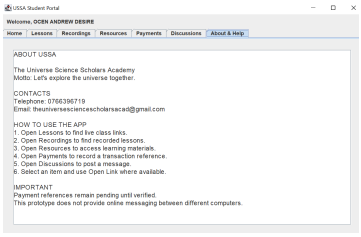
Learning Resources



Payments Section



Discussions



About & Help

11. Conclusion

The USSA Student Portal is an educational desktop application that enables students to access learning content, explore resources (library), record payment references, and have educational discussions.